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Paleomagnetic study of an arcuate fold belt developed on a marginal orogen: The Cajamarca deflection, northern Peru

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ABSTRACT

A paleomagnetic study of over 500 cores from 48 sites sampled in Albian to Turonian sedimentary formations along the Cajamarca deflection in northern Peru indicates that these lithologies carry a stable remanent magnetization. In the two major segments of the deflection, positive fold tests establish that the magnetization predates the folding, except for five sites in the western part which appear to be remagnetized by the intrusion of Paleocene granites. The eastern part of the deflection does not appear to have undergone any significant rotation, whilst a post-late Eocene counterclockwise rotation of about 25° is recorded in the central and western parts. This angular difference only accounts for about half of the difference in the strikes of structures in the two sections of the deflection. When considered with the available tectonic, sedimentological and geodynamic data, these results indicate that the arcuate shape of the Cajamarca fold belt results from complex processes. Early multi-phased compressional events of Paleocene–Eocene age created the folds and built an arcuate structure with non-rotational mechanisms. Then, in post-late-Eocene times the structure reached its present form after rotation of its central and western parts. The bearing of these results on the geodynamic evolution of the Andean Cordillera is discussed.

1. Introduction

The Cajamarca deflection is one of the major structural features of the Andean Cordillera of northern Peru. It is defined by a progressive change over a distance of 200 km in the strike of regional scale folds, between the Eastern Cordillera (Maranon geanticline) and the coastal area (Fig. 1 and 2). The eastern limb of the belt, south of 7°30′S, trends N160 along the edge of the eastern Cordillera and is parallel to the main structural, geomorphological and magmatic features of the Peruvian Andes and the associated oceanic trench. In contrast, the trend of the main structures in the Cajamarca transect between 7°S and 7°30′S is N110. North of 7°S, the trend of the western part of the belt is obscured by the overlying late Mesozoic and Cenozoic volcanic and plutonic belt.

In the geodynamic setting of the Andean marginal orogen, the origin of the well-defined N110 trending structures of the Cajamarca deflection is a rather puzzling problem. The northern NE–SW paleogeographic boundary of the Jurassic Chicama pull-apart basin may have played an important role in the present-day arcuate shape of the belt [1,2]. Recent radiometric data [3] yield some time constraints on the deformation of the deflection but no compelling evidence for the mechanism of its formation has yet been reported. In particular, it is not well established whether the arcuate shape of this belt is a primary tectonic feature or whether it originates at least partly from oroclinal bending of pre-existing folds.

The purpose of this study is to obtain reliable paleomagnetic results from a large number of sites distributed along the two major segments of the deflection in order to gain insight into the mechanisms and the age of formation of the structure.

2. Geological setting and paleomagnetic sampling

Between 6°30′S and 8°S, the Western Cordillera of northern Peru consists dominantly of

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well-exposed Mesozoic–lower Cenozoic sedimentary sequences and Cenozoic volcanic units. Large bodies of acid plutonic rocks (Andean batholith) outcrop along the coastal foothills.

The basement of the Mesozoic sequence is not exposed along the Cajamarca fold belt. Precambrian and/or Paleozoic units constitute most of the Maranon geanticline (Eastern Cordillera) and the Olmos massif located about 200 km northwest of Cajamarca (Fig. 2). Thus, some authors [1,4] assume that the Precambrian–Paleozoic basement is present at depth under the Mesozoic series.

Underlying the Cretaceous sequence, the oldest rocks exposed along the Cajamarca deflection are Middle to Late Jurassic sequences which have different characteristics north and south of an abrupt NE–SW boundary located near the 7°S parallel. North of 7°S, continental andesitic arc and volcanosedimentary units of Middle Jurassic to Early Cretaceous age (Colan Formation, Oyotun Formation of the Zana Group) underlie the Cretaceous sequence [5,6,7]. These formations are part of a NNE–SSW trending continental volcanic arc, which has also been reported in Ecuador and Colombia and is related to the southeastward convergence between an oceanic paleoplate and South America [8,9]. South of 7°S, the sedimentary formations of the Tithonian–Berriasian Chicama Group [10] record the opening of a deep turbiditic trough which has been interpreted as a back-arc pull-apart basin [11]. These significant differences in the Jurassic evolution recorded north and south of 7°S most probably reflect differences in the nature, mechanical properties and thickness of the crust. These differences are related to pre-Andean evolution of the crust and to segmentation of the subduction zone, which has been active since the early Mesozoic along the western South American margin [12,13,14].

The stratigraphy of the Cretaceous sedimentary sequence of the Cajamarca area is now quite well known [4]. It consists of Neocomian deltaic sandstones and Aptian–Santonian platform carbonates which are subdivided into eleven well-dated formations with specific facies and characteristic sedimentary sequences [2,4,15,16]. Facies and thickness variations of the Cretaceous formations indicate that sedimentation was controlled

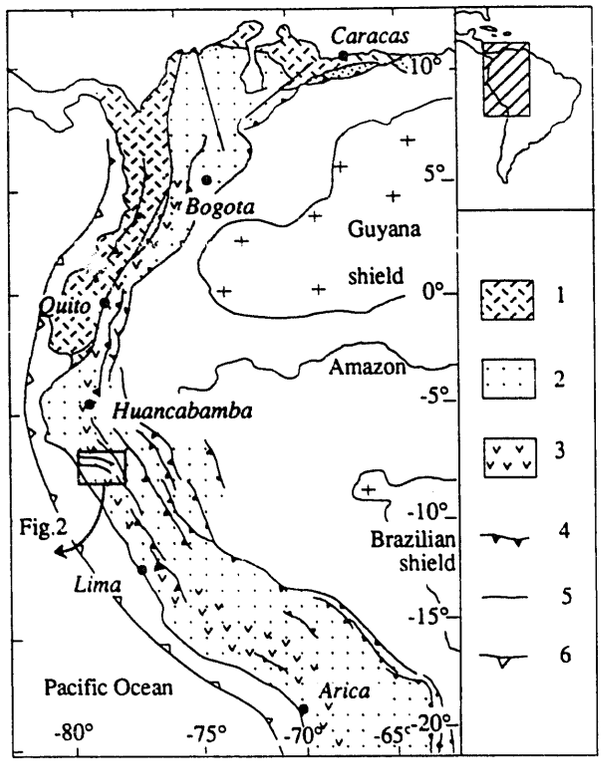


Fig. 1. Main geological characteristics of northwestern South America. *1* = accreted oceanic terranes (Northern Andes); *2* = Peruvian Central Andes; *3* = volcanic and plutonic Cenozoic belt; 4 = main intracontinental thrusts; 5 = main intracontinental faults; 6 = present-day oceanic trench.

mainly by paleogeographic features parallel to the margin but also by oblique relief in the Cajamarca region [16]. The Paleogene continental red beds of the Chota Formation [17] overlie the Cretaceous sequence on the eastern side of the Western Cordillera, while thick Cenozoic volcanic units unconformably overlie large areas of contrasting topography composed of Mesozoic rocks on its western side (Llama, Porculla and Huambos Formations) [3,5].

Recent radiometric dating of the unconformities separating the Cenozoic volcanic units and of tuffaceous layers interbedded in the Chota red beds [3,5] indicates that the structural features of the Cajamarca fold belt do not result from a single late Eocene tectonic phase as was previously believed [4,18,19]. In the western part, between Tembladera and Llama, the lower Llama volcanic sequence, which yielded a 54.8 Ma K/Ar

age, directly overlies tight E–W concentric folds with a marked unconformity and a basal conglomerate. In the eastern area, volcanic layers east of Cajamarca interbedded in the Chota red beds dated at about 50 and 44.2 My, are thus partly coeval with the Llama volcanics, and overlie the Cretaceous sequence with a slight unconformity. Thus, the western structures of the Cajamarca deflection result mainly from a poorly dated Late Cretaceous to early Cenozoic tectonic event, which is less marked in the east. In the central and eastern sectors, the Chota red beds are in turn clearly involved in the E–W trending folds of the Cajamarca area, and in the folds along the Eastern Cordillera (Marañon Thrust and Fold Belt; MTFB) [20]. These structures arise from the 43–42 Ma Incaic event [3,5] which also produced moderate folding of the Llama volcanics in the west. Therefore it appears that at least two major tectonic phases have given rise to the Cajamarca deflection.

Structural position and freshness of outcrop were the main criteria for the selection of paleomagnetic sampling sites. A portable spinner magnetometer was used in the field to select the most intensely magnetized formations among geologically equivalent sites. Lithologies with magnetizations lower than the noise level of this instrument ($\sim 5 \times 10^{-4}$ A m^{-1}) were sampled when it was impossible to locate more intensely magnetized outcrops. Most of the sampling (42 sites) was done on the black marly limestones of the middle Albian Pariatambo Formation, which is generally fresh and well-bedded. These fine-grained lithologies were chosen because their sedimentation rate is usually low enough to completely average out the secular variation of the geomagnetic field and also because they are typically deposited in low-energy environments. Six sites were sampled in other Cretaceous formations (sites AN02, 03 and 07 in the lower Albian Chulec Formation, AN 05 and 69 in the upper Albian–lower Cenomanian Yumagual Formation, and AN 57 in the Turonian Cajamarca Formation).

Cores were drilled using portable drilling equipment. Either an electric corer or a petrol-

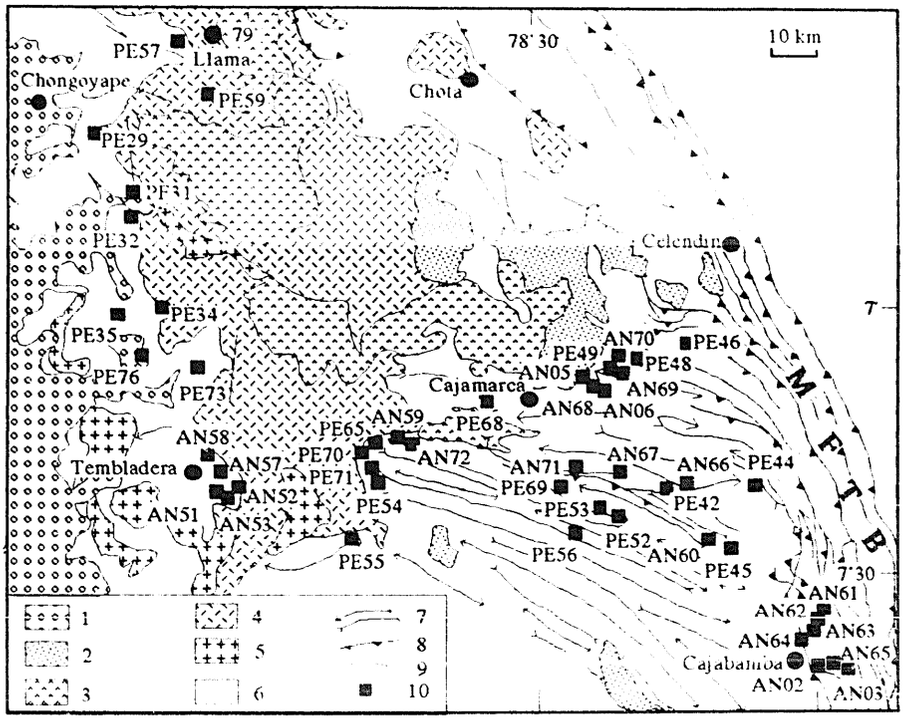


Fig. 2. Geological map of the Cajamarca deflection [after 5]. *1* = coastal desert; *2* = mio-pliocene volcanics; 3 = Huambos volcanics; *4* = Llama volcanics; *5* = intrusive stocks; *6* = undifferentiated Mesozoic formations; *7* = fold axes; *8* = thrusts; *9* = faults; *10* = sites. *MFTB* = Marañon Fold and Thrust Belt.

driven drill was used, depending on the nature of the outcrop. A minimum of eight cores were drilled from each site and each core was oriented with both a magnetic and a sun compass. The geographical and structural distribution of the 48 sampled sites is shown in fig. 2.

3. Results

3.1. Magnetic properties

Measurements of remanent magnetization were made using a LETI three-axis cryogenic magnetometer. The natural remanent magnetizations (NRM) ranged between $0.15 \cdot 10^{-3}$ and $5 \cdot 10^{-3}$ A m^{-1}. For many samples the thermal demagnetization behaviour is characterized by a rapid decrease in the NRM between 20°C and 100°C, followed by a more gradual decrease. In most cases, complete removal of the NRM was obtained at temperatures of between 400° and 500°C (Fig. 3a).

Stepwise acquisition of isothermal remanent magnetization (IRM), which was performed for at least one sample per site, indicates that in many cases 95% of the saturation IRM (SIRM) is acquired in fields of less than 0.2 T, while in others saturation of the IRM is not observed even at 2T (Fig. 3b). Subsequent alternating field (AF) and thermal demagnetization of the SIRM shows differences in coercivities and blocking temperatures for samples from different sites (Fig. 3c and d). These differences reflect the presence of different magnetic minerals.

Attempts to characterize the magnetic minerals by thermomagnetic analyses were generally unsuccessful because it proved difficult to obtain magnetic separates from the Pariatambo Formation. The only separate obtained (site AN 60) was heated to 700°C and cooled in a nitrogen atmosphere. The induced magnetization decreases progressively up to 400°C, then increases between 400°C and 500°C before dropping to noise level at about 600°C (Fig. 4a). This thermomagnetic behaviour is not reversible upon cooling. When a second heating is performed on the same sample, a continuous decrease in the induced magnetization is observed at about 580°C, where no magnetization remains (Fig. 4b). This behaviour indicates the presence, in addition to magnetite, of a

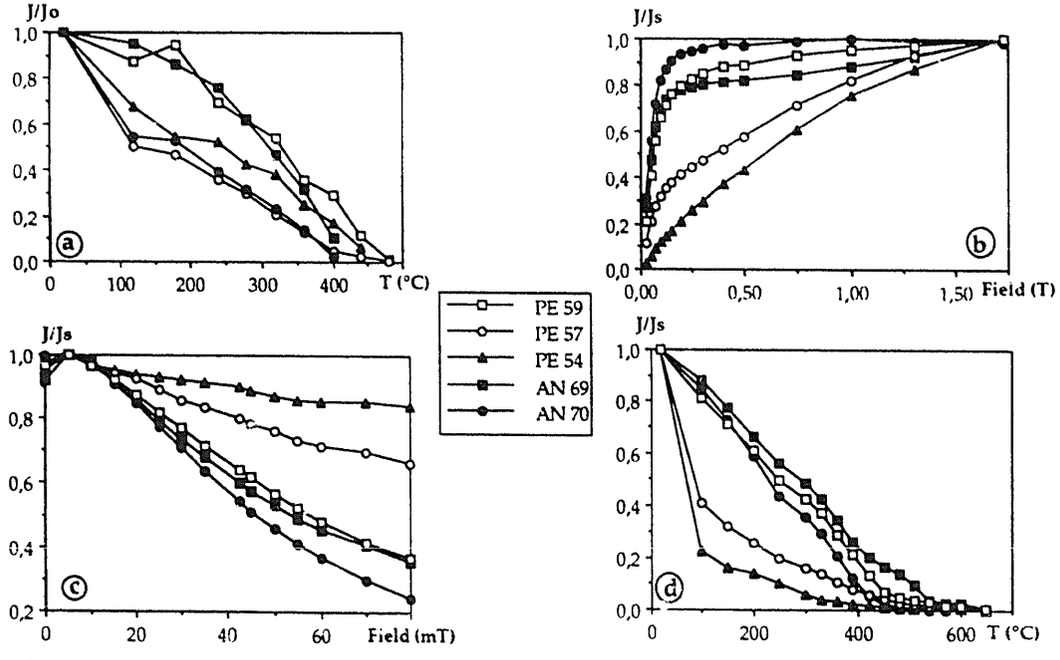


Fig. 3. (a) Thermal demagnetization curves, normalized to the natural remanent magnetization (NRM). (b) Normalized isothermal remanent magnetization (IRM) acquisition curves. (c) Normalized curves of the subsequent alternating field demagnetization of the maximum IRM. (d) Normalized curves of the subsequent thermal demagnetization of the maximum IRM.

TABLE 1

Reliable paleomagnetic results from the Cretaceous sedimentary formations of the Cajamarca deflection

Sites	n/N	Before B.C.		After B.C.		K	$\alpha_{95}(°)$	Bedding attitude
		$D(°)$	$I(°)$	$D(°)$	$I(°)$			
(a) Eastern part of the deflection								
AN02	10/10	120.0	22.2	158.0	23.7	99	4.4	140,75
AN03	7/9	316.2	−30.4	348.7	−10.7	259	3.3	140,75
AN61	13/13	5.2	−18.0	350.6	−26.0	163	3.0	345,35
AN62	9/12	13.6	−16.0	351.2	−26.0	873	1.6	352,54
AN63	6/8	15.4	−5.0	351.5	−31.0	213	3.8	345,70
AN64	6/6	317.0	−41.0	348.2	−26.0	23	11.8	135,50
AN65	10/13	307.0	−19.0	336.0	−24.0	508	2.0	145,65
Mean (after B.C.)	$N = 7/7$			$D = 346°$	$I = -24°$	$K = 88$	$\alpha_{95} = 6.5°$	
$R = -4.9° \pm 6.2°$; $F = 5.6° \pm 7.2°$								
(b) Central part of the deflection								
AN05	8/10	332.0	18.5	320.8	−20.6	105	4.8	290,70
AN06	4/4	332.6	26.6	334.3	−23.0	241	4.5	285,70
AN51 *	11/12	334.7	−25.0	338.3	8.5	25	8.4	101,40
AN52 *	8/9	341.4	−26.0	345.6	9.0	42	7.6	110,45
AN53 *	8/11	311.2	−29.0	323.4	−7.0	168	3.8	110,43
AN57 *	9/11	139.2	11.4	123.5	10.6	223	3.0	312,70
AN59	10/10	348.5	5.0	335.8	−30.0	107	4.2	305,58
AN60	11/12	277.0	−24.0	293.0	−17.0	274	2.5	095,42
AN68	14/14	326.0	8.0	314.4	−35.0	208	2.6	278,60
AN69	14/14	322.7	2.7	314.3	−25.0	235	2.4	285,48
AN70	11/12	327.6	−23.0	303.3	−28.0	537	1.8	310,48
AN71	10/10	285.8	−24.0	303.8	−16.0	76	5.1	105,50
AN72	10/19	338.0	4.0	335.5	−20.0	331	2.0	variable
PE29	11/12	336.0	−16.0	336.0	−16.0	365	2.2	horizontal
PE32	8/10	165.0	28.3	139.1	27.6	141	4.2	332,48
PE34 *	9/12	156.0	32.2	158.0	−21.2	18	10.9	086,56
PE35	5/10	324.6	−55.0	323.3	−35.0	50	8.8	050,20
PE42	8/9	337.4	−38.8	331.8	−30.2	32	8.8	variable
PE45 **	10/11	260.0	−30.0	242.8	−17.0	78	5.0	270,40
PE46	8/10	331.0	−60.0	340.0	−12.0	43	7.5	080,50
PE48	10/10	330.4	−18.6	312.1	−30.4	338	2.4	305,40
PE49	6/7	328.8	−13.7	324.0	−20.5	63	7.2	300,15
PE52	10/11	90.0	33.0	130.0	40.0	28	8.3	118,50
PE54	13/13	321.0	−30.0	340.0	−25.0	126	3.3	144,36
PE56 **	9/10	167.0	−23.4	138.0	−55.6	81	4.9	110,45
PE57	10/10	333.0	−24.0	318.1	−42.4	31	8.4	289,32
PE59	8/8	337.0	−18.0	331.0	−33.0	98	5.0	275,19
PE65	7/9	289.0	−18.0	304.0	−12.0	106	5.1	111,50
PE68	7/9	161.6	19.2	159.5	23.7	50	7.5	300,07
PE69	11/12	244.5	−9.5	329.1	−33.1	231	2.7	112,120
PE71	10/11	320.0	−8.7	323.7	−32.0	29.5	8.1	210,25
PE73	10/10	351.1	−7.0	349.3	−14.0	74	5.2	305,12
PE76	11/12	352.6	−2.3	354.8	−28.0	63	5.3	246,27
Mean (after B.C.)	$N = 31/33$			$D = 324.7°$	$I = -26.4°$	$K = 27.3$	$\alpha_{95} = 5.0°$	
$R = -26.2° \pm 5.2°$; $F = 8.0° \pm 6.5°$								

* Sites remagnetized after the deformation (see text for explanation)

** Sites diverging by more than 2 standard deviations from the mean and rejected in the statistic

small amount of a magnetic mineral which is unstable at temperatures higher than 400°C and evolves into magnetite during heating. Because of the close similarity between this thermomagnetic behaviour and other independent observations [21,22], we suggest that this mineral is an iron sulphide. The results of thermal and AF demagnetizations described below indicate that magnetite is the main magnetic carrier in the majority of the samples. The higher coercivities observed at some sites suggest the presence of haematite or goethite.

3.2. Paleomagnetic results

Thermal and AF demagnetizations isolated the same stable component of magnetization in different specimens of the same cores. The thermal method was preferred because, overall, it yielded more consistent results. All sample were stepwise demagnetized in twelve–fifteen steps from room temperature to the limit of reproducible results, which ranges between 400° and 500°C. No significant changes in the magnetic susceptibility, which was routinely measured at each step with a susceptibility bridge, were observed up to the highest temperatures. This indicates that, for most of the sites, the magnetic properties of the samples did not change significantly during demagnetization, in agreement with the magnetic mineralogy analyses. Stable paleomagnetic directions, isolated after heating at 200–250°C, were determined using orthogonal demagnetization diagrams, some of which are shown in Fig. 5.

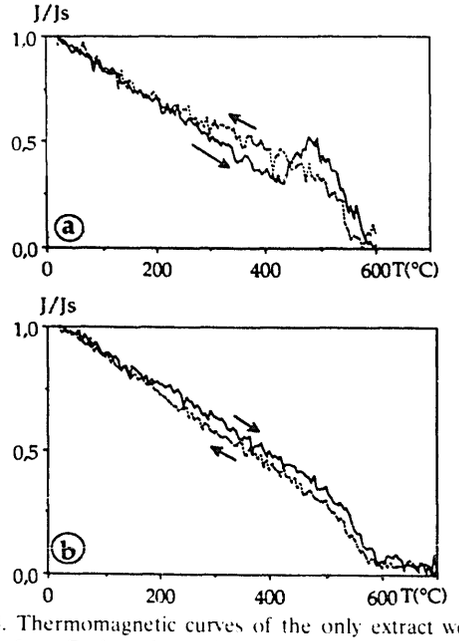


Fig. 4. Thermomagnetic curves of the only extract we could obtain from Pariatambo sediment (see text). (a) First heating (black line) and cooling (dashed line) in a nitrogen atmosphere. (b) Second heating and cooling curves (see text for explanation).

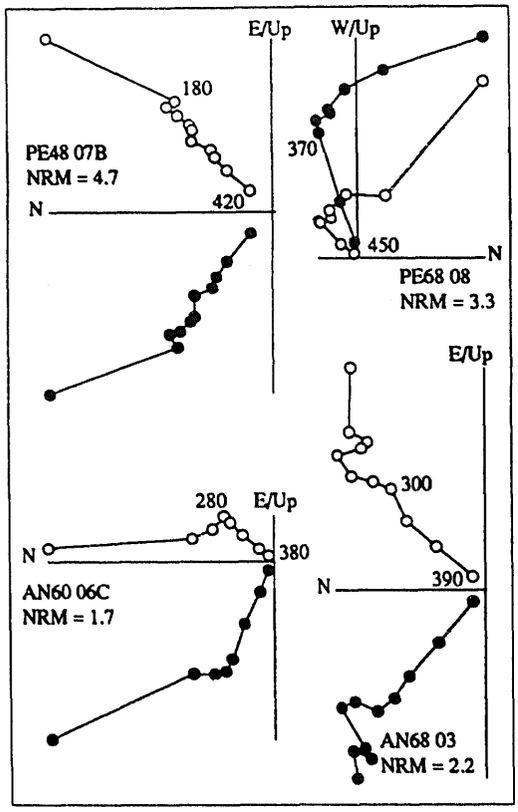


Fig. 5. Orthogonal projection diagrams of the thermal demagnetization of samples from the Pariatambo formation after tilt correction. Dots and circles correspond to projections on the horizontal plane and on the N–S vertical plane, respectively. The NRM values are reported in 10^{-3} A m^{-1}. Some of the temperatures are indicated close to their corresponding points.

In 40 out of the 48 sites sampled, the characteristic remanent magnetization was determined with good accuracy: α_{95} is usually of the order of 5° at each site. These results are reported in Table 1 and Fig. 6, where, for the reasons discussed below, we have separated the results into those from the western-central and eastern parts of the deflection.

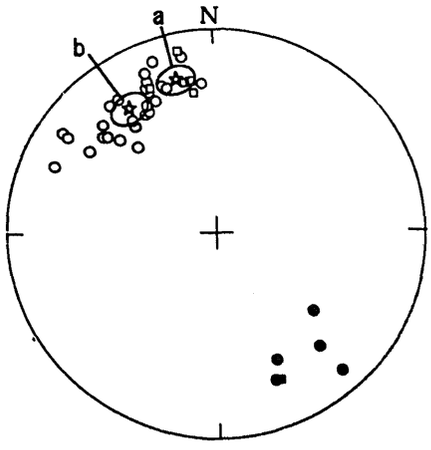


Fig. 6. Equal-area stereographic projection of the paleomagnetic vectors for the sampled formations. Open symbols correspond to upward vectors and closed symbols to downward vectors. Squares correspond to the eastern part of the deflection and circles to the western and central parts. A and B are, respectively, mean paleomagnetic directions with 95% confidence circles for the eastern and the western-central part of the deflection.

Data from the remaining eight sites were discarded for various reasons: For five sites, accurate determination of the stable direction could not be obtained because of erratic magnetic behaviour during thermal demagnetization (sites AN58, PE31, PE44, PE55 and PE70). For the three others, despite the almost rectilinear diagrams obtained from single samples the intra site scatter was unreasonably high, with a Fisher dispersion parameter k of less than 10 (sites AN66, AN67 and PE53).

3.3. Age of the magnetization

It can be seen from Table 1 that six sites have a reversed magnetization. Most of these sites (five out of the six) are paleontologically dated as Albian, so that, in principle, they should have been deposited during the long Cretaceous normal polarity interval. Possible explanations for the observed reversed polarities are (a) errors in the dating (reversed polarities would be expected for sites only slightly older), (b) occurrence of a short reversed interval during Albian times, as recently suggested by [23], or (c) local remagnetizations.

Two of the reversed polarity sites (AN57 and PE34), together with three normal polarity sites (AN51, AN52 and AN53), were sampled near the village of Tembladera on the western limb of the deflection, in the vicinity of intrusive stocks and thick volcanic deposits of Paleocene and Eocene age [3] (Fig. 2). The paleomagnetic directions observed at these sites after bedding correction are inconsistent, with north-seeking directions with upward inclinations (or conversely, south-seeking directions with downward inclinations), which one would, normally expect from southern hemisphere sites. It can be seen from Table 1, however, that the uncorrected directions are not significantly different from those observed after bedding correction in the central, N110-trending, part of the deflection. In our opinion, these sites were remagnetized during the intrusion of the Paleocene granites and they record the same

TABLE 2

Fold test obtained from the eastern and the central part of the deflection using the method given by McFadden and Jones [25]

N_a	N_b	$AR1$	$AR2$	B
A, Eastern part, N = 7				
3	4	9.136	0.599	0.820
B, Central part, N = 29 (31 sites, 1 horizontal, 1 variable bedding attitude)				
17	12	0.601	0.103	0.117
C, Central part, N = 24 (29 sites, 5 remagnetized)				
16	8	1.03	0.072	0.145
D, Central part, N = 29 (bedding-uncorrected directions of the 5 remagnetized sites considered with the bedding-corrected directions of the other sites)				
17	12	0.456	0.013	0.117

N_a and N_b are the number of directions in each flank of the megastructures. $AR1$ and $AR2$ are the first terms of eq. (7) of McFadden and Jones calculated before and after bedding correction respectively. B is the second term of the same equation at the 5% level of significance. The populations have a common true direction after bedding correction (positive fold test) when $AR2$ is lower than B.

counterclockwise declination as the E–W sites with respect to the present geomagnetic field direction. This implies that the westerly declinations of the Cretaceous formations all record a post-Paleocene rotation. This is consistent with the paleomagnetic results previously obtained [24] from younger formations in the same region. This point is further discussed below.

The bedding-corrected paleomagnetic directions of the four other reversed polarity sites (AN02, PE32, PE52 and PE68) are counterclockwise rotated with respect to the dipole field direction and are virtually antipodal to the normal polarity directions in the same region. Furthermore, considering that both the normal and reversed sites belong to a N110- and a N150-trending megastructures, we have performed the fold test proposed by McFadden and Jones [25]. The results, reported in Table 2, show that the test is positive at the 95% significance level. In the central branch it is worth noting that although the test is always positive its significance level is improved when the five remagnetized sites are not considered (Table 2, *C*); the significance is even greater when the directions of these sites before bedding correction are considered with the bedding-corrected directions from the other sites (Table 2, *D*). The fold test thus establishes that, in a general way, the magnetization predates the folding and supports the hypothesis of remagnetization of the five sites discussed above. In the interpretation of the data, we have thus considered these five sites with their in-situ directions.

In summary, the 40 sites reported in Table 1 are characterized by a stable characteristic remanent magnetization and can be used for structural interpretation. When the results are averaged using Fisher's statistics, a paleomagnetic direction of ($D = 324.7°$, $I = -26.4°$, $N = 31/33$, $k = 27.3$, $\alpha_{95} = 5.0°$) is obtained from the E–W trending central part of the deflection, while the eastern (Andean-trending) limb is characterized by a direction of ($D = 346°$, $I = -24°$, $N = 7/7$, $k = 88$, $\alpha_{95} = 6.5°$).

To test whether these two directions are significantly different, we first considered the ratio of the precision parameters k in the two parts of the deflection. The k_1/k_2 ratio observed ($k_1/k_2 = 3.22$) is slightly higher than the critical value for $F(F(60/12) = 2.38)$, so the two populations have statistically different precision parameters. Following McFadden and Lowes [25], we used the ratio $k_1/\gamma k_2$, where $\gamma = 1.35$ was determined so that the F test of $k_1/\gamma k_2$ is significant at the 95% level. The corresponding test of the mean directions [26] was then found to be positive at this same level ($0.348 > 0.087$). The paleomagnetic directions determined for the two parts of the deflection thus appear to be significantly different at the 95% statistical level.

4. Discussion

One of the major difficulties in the interpretation of paleomagnetic data from stable South America (SOAM) arises from the fact that the apparent polar wander path (APWP) for this continent is still poorly defined, in particular for the Cretaceous epoch considered here.

In this paper, we use the Upper Cretaceous–Paleocene pole recently proposed for SOAM [27]. These authors averaged the Upper Cretaceous–Paleocene poles recently determined from southeastern Brazil [28] and from southern Chile and Argentina [29], obtaining a pole defined by (80.7°N; 175.8°E; $A_{95} = 3.6°$). The validity of this pole was checked by transferring the paleopoles from other continents to South America using the kinematics of the opening of the Atlantic Ocean [30]. As the distribution of the 80 M.y. and 100 M.y. transferred poles is closely scattered about this new pole [27], the results obtained from Albian, Cenomanian and Turonian formations can all be referred to this pole without significant loss of accuracy. Assuming a geocentric dipole field, this pole corresponds to a paleomagnetic direction defined by ($D = 351°$, $I = -18°$) for the region studied here. The directions of the five remagnetized sites were referred to the available 40 M.y. pole [24] corresponding to the age of the remagnetization. The flattening and rotation parameters calculated from these data [31,32] (Table 1) yield new information on the geological evolution of the Cajamarca area and also, more generally, on the evolution of the Andean Cordillera.

The values of the flattening parameters, $F = 5.6° \pm 7.2°$ and $F = 8° \pm 6.5°$, for the western-central and the eastern sectors respectively, would

suggest some northward displacement of the Cajamarca area with respect to the stable craton, but they are not significant. The rather high uncertainty of these parameters arises to a large extent from a corresponding uncertainty of the reference pole for SOAM and no definitive conclusion can be drawn for the latitudinal movement with respect to the stable continent. We have then explored the possibility of relative movements in the coastal regions by comparing the mean Cretaceous paleolatitude (pL) of the Cajamarca area ($pLC = -13.2° \pm 3.5°$) with that of coeval formations of the Lancones Basin, some 300 km to the north ($pLL = -8.5° \pm 4.4°$) [33]. The difference, $\Delta pL = 4.7° \pm 4°$, which is independent of any reference pole, is slightly larger than the present-day latitudinal difference between the two zones ($\Delta L = 2.75°$). However, it is not really significant, so the only conclusion is that if some shortening has occurred in this region [2] its amplitude is smaller than the uncertainty of the paleomagnetic data.

More conclusive results are obtained from the analysis of the paleomagnetic declinations. The rotation parameter R calculated for the eastern part of the deflection ($R = -4.9° \pm 6.2°$) is significantly different from that of the central part ($R = -26.2° \pm 5.2°$). Firstly, this indicates that the eastern part of the deflection has not undergone any significant rotation with respect to SOAM since at least Albian time. Secondly, it indicates that the difference between the two R parameters for the central and the eastern limbs is of the order of 25°, and is independent of the choice of the reference pole. Thus, the central part of the deflection has undergone a counterclockwise rotation of about 25° with respect to the eastern, unrotated limb of the deflection. This rotation only accounts for about half of the present-day angular deviation (50°) between the trends of the eastern and the central segments of the deflection.

Thus, together with the radiometric data, these results suggest that the present orientation of the Cajamarca deflection results from a complex process: first a multi-phased tectonism of Late Cretaceous–Early Cenozoic age is responsible for the folding [3,5], then building up an arcuate structure with a non-rotational mechanism, and second the structure reached its present form after 25° counterclockwise rotation of its central and western parts.

In a previous article [24] we have shown that in this zone the slightly tilted Eocene Llama volcanic formation and the undeformed late Eocene Huambos Formation, which overlie the Cretaceous formations sampled here, and the intrusive formations dated in this area at about 43 M.y. [34], have all undergone a post-late Eocene counterclockwise rotation of 19° ($R = -19° \pm 9.7°$). This value is almost identical to that determined here for the mid-Cretaceous formations. Thus, the counterclockwise rotation of the central and western parts of Cajamarca deflection reflects the pervasive post-late Eocene counterclockwise rotation of the Peruvian margin and is not related to the tectonic mechanisms responsible for the building of the deflection. The slight difference between the two R parameter values may possibly result from small regional counterclockwise rotations (about 5°) between Albian and late Eocene time or, more realistically, from differences and uncertainties in the position of the SOAM reference poles for these two epochs.

The interpretation of a rotated western-central segment of the Cajamarca deflection and an unrotated eastern segment requires that a "decoupling" mechanism exists between these two sections. Evidence for such a decoupling is offered by the late activity of the Marañon Fold and Thrust Belt (MTFB), the main emplacement of which took place in the late Eocene ($\sim$ 42 M.y.). Younger movements along dextral strike-slip faults trending N160 and located east and northeast of Cajabamba (Rio Crisnejas) have also been observed [2], and Mourier has suggested that these may have played the role of lateral ramp during the emplacement of the MTFB [5]. Strong evidence of the possibility of relative differential movements between the two segments of the deflection is thus yielded. Detailed structural analysis of this strike-slip zone could yield definitive results on this point, but this is unfortunately not possible at present.

5. Conclusions

Two main conclusions may be drawn from the results reported above.

First, the Andean-trending structures in the eastern part of the Cajamarca deflection do not appear to have undergone any significant rotation. Because of the consistency of the results from the coast to the Altiplano in central Peru with those from Cenozoic formations in northern Peru and southern Ecuador [24,27], we have suggested that the Peruvian margin has rotated as a semi-rigid body rather than by rotation of individual blocks [35]. However, because no sites could be sampled in the Eastern Cordillera, the results did not yield any constraints on the eastern limit of the rotation. The results obtained here appear to indicate that, at a local scale in the Cajamarca area, the western boundary of the Eastern Cordillera may constitute this eastern limit. This boundary may well have acted as a buttress to rotations, as suggested by the eastward shift of deformation through the Eastern Cordillera between the Incaic (Eocene) and the Quechua (Miocene) tectonic phases [3,5]. We have, however, no evidence that the Eastern Cordillera represents the eastern limit of the rotation everywhere along the Peruvian margin. North of Cajamarca, for instance, in the Bagua region, where both the Olmos and the Marañon massifs plunge under the Mesozoic sedimentary cover, there is no evidence of such a buttress. A counterclockwise rotation of about 10° is reported from the Cenomanian Yumagal Formation in this region (5°54′S, 78°09′W) [36]. Although no reliable conclusion can be drawn from a single site, it may nevertheless indicate that the position of the eastern limit of the area affected by the rotation varies as we pass from northern to southern Peru.

Second, our results demonstrate that the rotation of the Peruvian margin has contributed to, but has not created, the curvature of the Cajamarca deflection. This rotation is unrelated to the folding, and a non-rotational mechanism has played about half a role in the arcuate shape of the deflection.

Marshak [37] recently demonstrated that similar arcuate structures may be produced by several rotational or non-rotational mechanisms. The available structural data for the Cajamarca deflection are not detailed enough to constrain one or the other of the suggested mechanisms. However, the paleomagnetic results reported here indicate that half of the deflection formed without rotation. It is therefore possible to suggest a reasonable scenario for the formation of the Cajamarca deflection.

As described above, the boundary of the Jurassic Chicama Basin corresponds, eastward, to the N160-trending western edge of the Marañon geanticline and, at the latitude of Cajamarca, to a sharp NE–SW paleogeographic boundary characterized by a major change in the Mesozoic facies distribution. These geological boundaries determine the geographic distribution of the Chicama Formation, which represents a décollement layer [5,11], and could in turn affect the initial geometry of thrust faults and associated hanging wall structures [38]. Analogue sandbox model experiments indicate that thrust geometry is directly correlated to the spatial extent of the décollement layer [39]. In the case of arcuate geometry, such as is observed in the Cajamarca area, the orogen will initiate as a non-rotational arc [37,39], exactly as shown by the paleomagnetic results. Such non-rotational arcs that form along irregular margins as a consequence of curved faults may be to some degree analogous to arcs resulting from "moulding" of orogens onto the irregular fringes of older cratons [37,40,41].

Thus, it is not necessary to invoke a radial stress field [5] to account for the arcuate shape of the deflection. Rather, all the geological, paleomagnetic and analogue model data are consistent with the hypothesis that the arcuate Cajamarca deflection results from the closure of the filled pull-apart Chicama Basin between Early Cretaceous (Albian) and Eocene times [1,33; Jaillard, pers. commun.]. This closure results in the non-rotational building of curved folds and thrusts with deflections of about 20°–30°. The final curvature of the deflection was then acquired as a result of a post-late Eocene regional rotation which has affected the Western Cordillera and the coastal region in this area.

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